

"Hello Everyone!"

Maths: Prime Numbers



* Count of factors $N \rightarrow O(\sqrt{N}) \rightarrow \underline{\underline{TC}}$
 $O(1) \rightarrow \underline{\underline{SC}}$

Prime Number :-

↳ Number having exactly 2 factors ✓

1 and itself.

(+ve)
numbers

2, 3, 5, 7, 11, 13, 17, 19, ...

Smallest
prime
number

-1 < 1

(Q.1) Given an int n , check every number
from 1 to N , if it is prime or Not

eg:-

$N=10$

[1 2 3 4 5 6 7 8 9 10]

Ans

isP

[F T T F T F T F F F]

BF:- ①

$\forall 1 \rightarrow n,$

Count # factors.

if factors == 2

else

T

F

$O(\sqrt{n})$

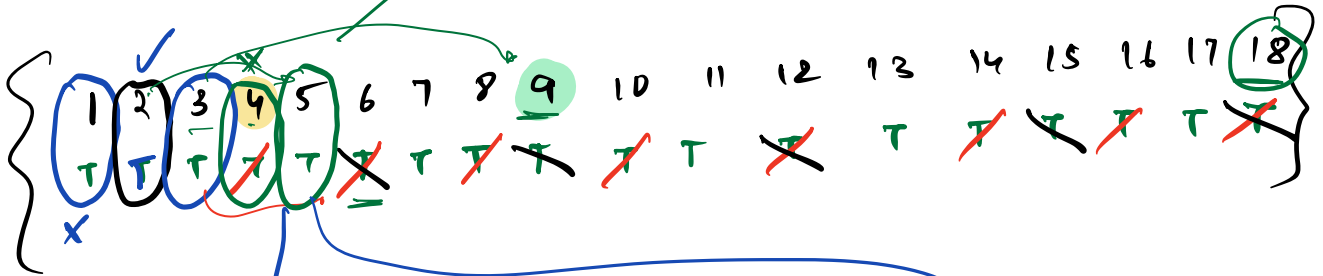
TC: $O(N + \sqrt{N})$

SC: $O(1)$

Sieve

N=18

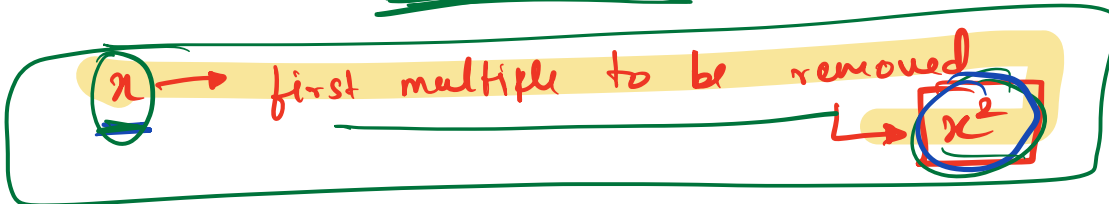
10 15 20 . . .



what is the first no. that 5 will help getting removed.

- ✓ 10 → 2 * 5
- ✓ 15 → 3 * 5
- ✓ 20 → 2 * 2 * 5
- 25 → 5 * 5

stop



$i * i > N \rightarrow$ stop

$\forall i, \text{isP}[i] \rightarrow T$

$\text{isP}[0] = \text{isP}[1] = F$

$\text{isP}[i] \rightarrow T, \text{ } i \text{ is Prime}$

for ($i = 2$; $i * i \leq N$, $i++$)

{

if ($\text{isP}[i]$)

{

for ($j = i * i$; $j \leq N$; $j = j + i$)

{

$\text{isP}[j] = \text{False}$

}

}

}

return isP

3
↓
6
↓
9

$i = 4$
↓
6
↓
8
↓
10
↓
12

all mul of i

$N = 5$

$i = 2$
 $j = 4$ $\rightarrow$ ≤ 5 , $j = j + 2$

~~0~~
~~F~~

~~1~~
~~F~~

2
T

3
T

~~4~~
~~F~~

5
T $\rightarrow$

$$\underline{i = 3}$$

$$\underline{j = 9}$$

i	j	# iterations
2	4, 6, 8, ...	$\underline{N/2}$
3	9, 12, 15, ...	$\underline{N/3}$
4	<u>0</u>	<u>0</u>
5	25, 30, 35, ...	$\approx N/5$
...		

$\left(\frac{N}{2} + \frac{N}{3} + \frac{N}{5} + \frac{N}{7} + \frac{N}{11} + \dots \right) < N \left(\frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \frac{1}{5} + \frac{1}{6} + \dots \right)$

Sum of reciprocals of primes

Sum of reciprocals of all nos.

$\sum_{i=1}^{\infty} \frac{1}{n} \Rightarrow \int \frac{1}{x} dx = \log(x)$

$$\frac{N^*(\log(\log N))}{\text{very few}}$$

$$\leq \frac{N * \log N}{\text{TC}}$$

$$O(N * \sqrt{N})$$

$$TC: O(N * \log(\log N))$$

$$SC: O(1)$$

(Q) Given an int N ,
for all the numbers from
1 to N ,

Count the factors of ^{all} the number.

eg:-

$N=8$

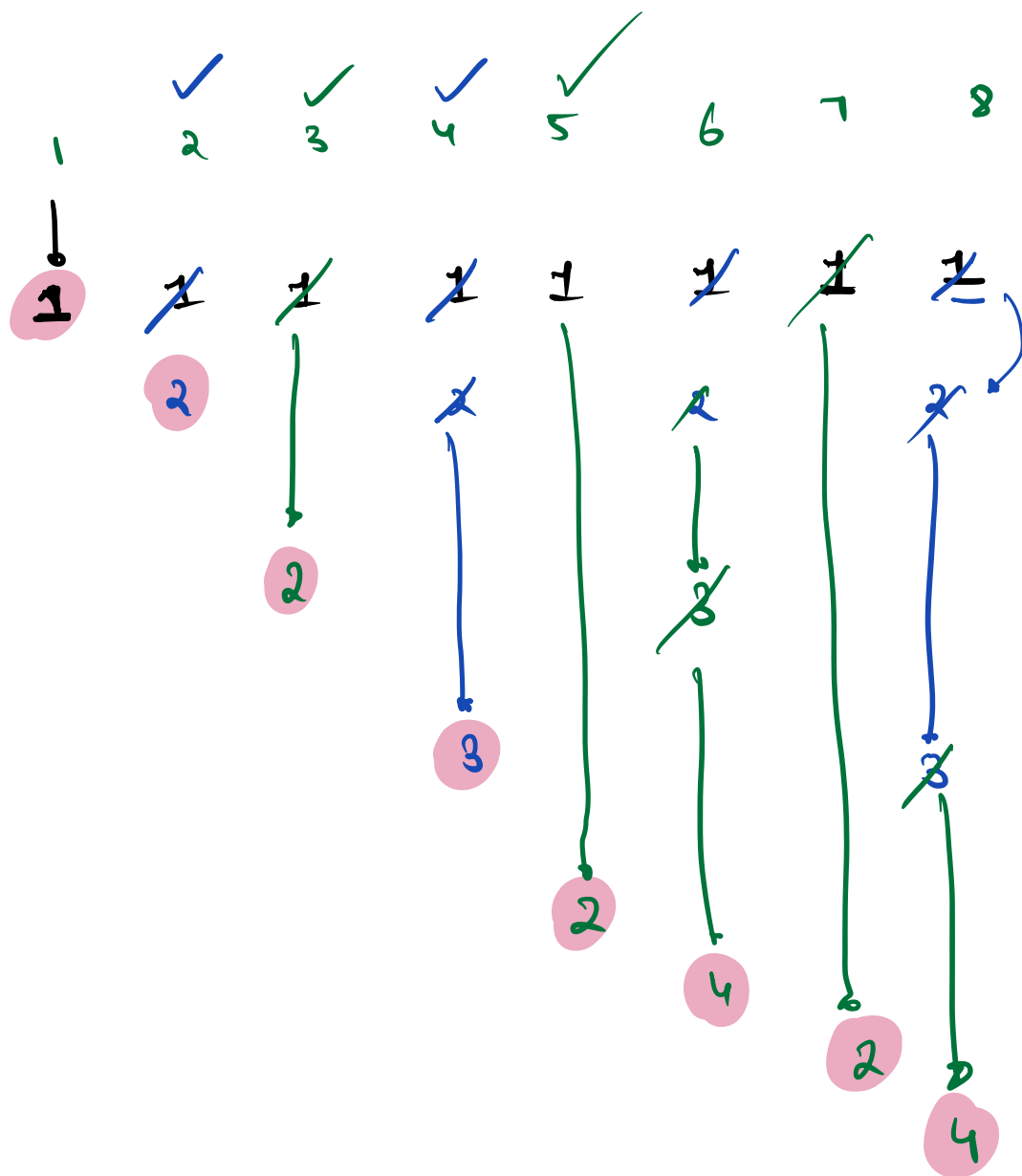
Ans:

	1	2	3	4	5	6	7	8
	1	1	1	2	2	2	2	4
	1	2	2	3	2	4	2	4

BF:- $\forall$ 1 to N ,

count factors separately

→ TC: $O(N \cdot \sqrt{N})$



Logic:

n will give first chocolate to n .

then to all the multiples of n .

count of factors
↑
+i, $\text{cof}[i] = 0$

for ($i=1$; $i \leq N$; $i++$) → go to all nos

{
for ($j=i$; $j \leq N$; $j = j+i$)

{

$\text{cof}[j] += 1$

}

}

return

cof

→ i is dist'n
chose to j

TC: $O(N * \log N)$

SC: $O(1)$ ★

10:12

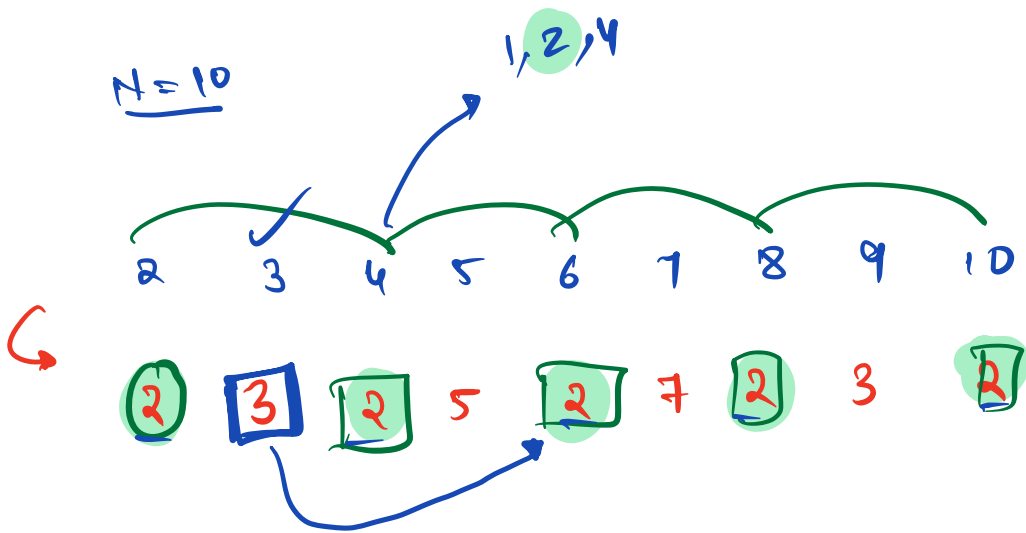
(8) Given an int N ,

find the smallest Prime factor

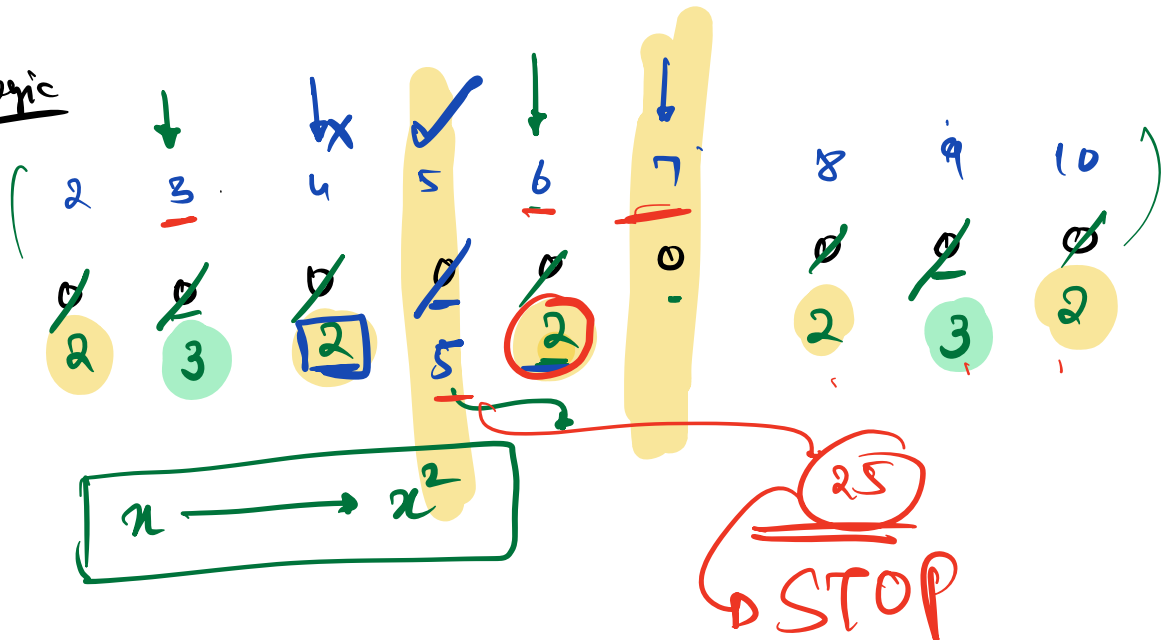
for all the nos from 1 to N .

given: $N \geq 2$

SPF



Logic



$\forall i, \text{SPF}[i] = 0$

for ($i = 2$; $i \leq N$; $i++$)

{

if ($\text{SPF}[i] == 0$) // i is a prime no.

{

$\text{SPF}[i] = i$

for ($j = i * i$; $j \leq N$; $j = j + i$)

{

if ($\text{SPF}[j] == 0$) eq of 6

{
1 $\text{SPF}[j] = i$
3

}

}

for ($i = 2$; $i \leq N$; $i++$)

{

if ($\text{SPF}[i] == 0$)

{

$\text{SPF}[i] = i$

}

}

part 1

part 2

return spr

TC: $O(\underline{N * \log(\log N)} + \underline{N})$

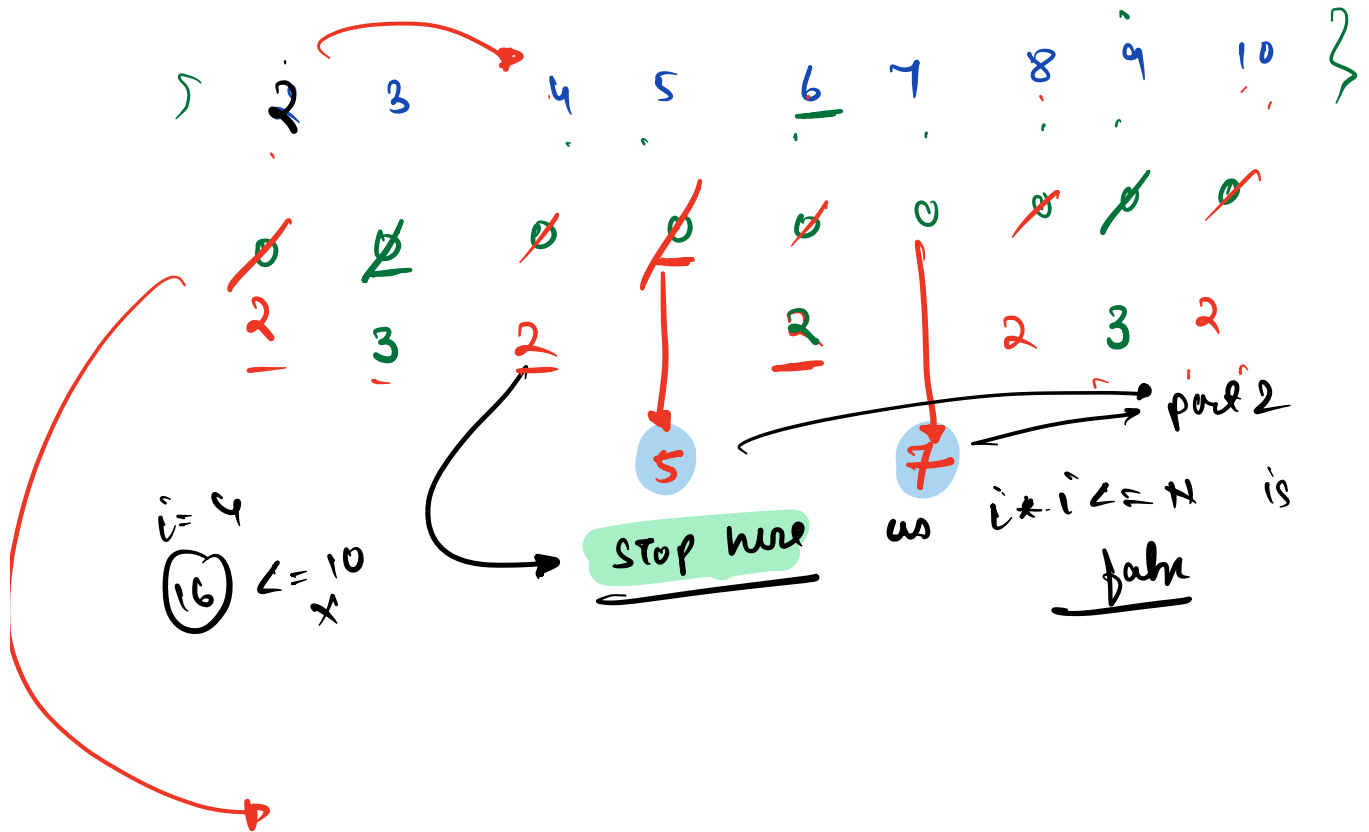
part 1

$\approx O(N \log(\log(N)))$

SC: O(1)

Dry Run

$x \rightarrow x^2$



(Q)

Count no. of factors of N ,

given the $\text{spf}[i] \rightarrow \forall i [1 \text{ to } N]$

eg:
① 45 $\rightarrow$ 6 (1, 3, 5, 9, 15, 45)

② 32 $\rightarrow$ 6

1	2	4	8	16	32
$\swarrow$	$\downarrow$	$\downarrow$	$\downarrow$	$\downarrow$	$\downarrow$
2^0	2^1	2^2	2^3	2^4	2^5

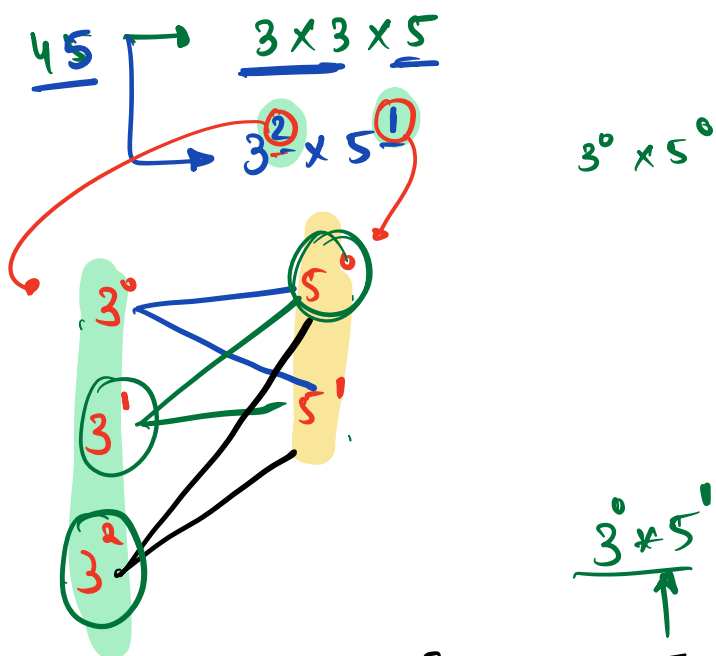
32 $\rightarrow$ $\left(\begin{smallmatrix} 5 \\ 2 \end{smallmatrix} \right) [0 \rightarrow 5]$

of factors of $2^k = \{2^0, 2^1, 2^2, \dots, 2^k\} \rightarrow \boxed{\{k+1\}}$

$$45 \rightarrow 9 \times 5$$

$$\quad \quad \quad \swarrow$$

$$\quad \quad \quad 3 \times 3$$



$$45 \rightarrow [1, 3, 5, 9, 15, 45]$$

$$\begin{array}{c} 3^0 \times 5^0 \\ \uparrow \\ 3^0 \times 5^1 \\ \uparrow \\ 3^1 \times 5^0 \\ \uparrow \\ 3^2 \times 5^0 \\ \uparrow \\ 3^2 \times 5^1 \end{array}$$

factors of $3^x 5^y \rightarrow$

$$(x+1) * (y+1)$$

factors of $2^a + 3^b + 5^c \rightarrow$

$\rightarrow (a+1) * (b+1) * (c+1)$

eg: $N = 490 \rightarrow$ find (Q) No. of factors of N

490×10

$$7 \times 7 \times 5 \times 2$$

$2! \times 2$

$2 \times 5 \times 7 \times 7$

$2^1 \times 5^1 \times 7^2$

$$\Rightarrow (1+1) * (1+1) * (2+1)$$

$$= 2 \times 2 \times 3$$

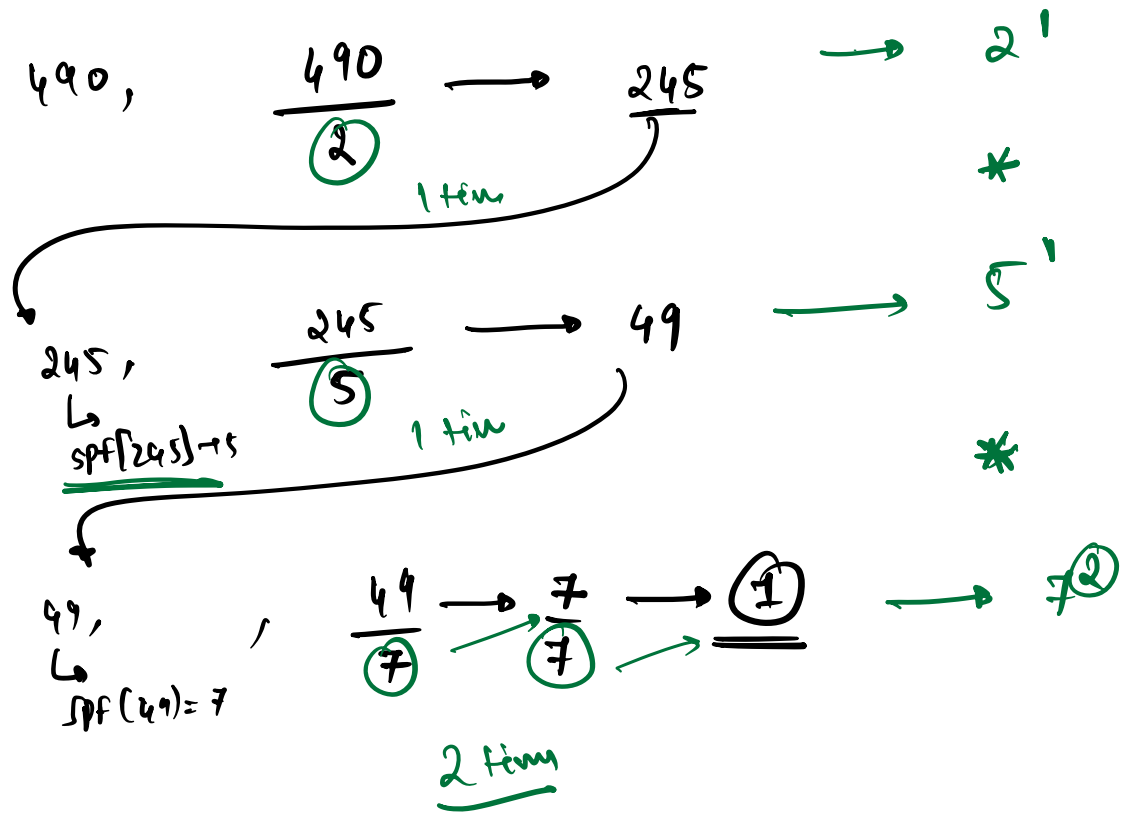
$$= 12$$

4×3

$= 12$

490

spf[490] $\rightarrow 2$



```

t = 1
while (N > 1)
{
    s = SPF[N]
    pow = 0
    while (N % s == 0)
    {
        N = N / s
        pow++
    }
    f = t * (pow + 1)
}
return f

```

Smallest Prime factor of n

$$\frac{690}{2} \rightarrow \underline{265}$$

2'

$$f = 1 \times (1+1) \\ = 2$$

$$\swarrow \\ f = 2 \times (1+1) \\ = 4$$

$$f = 4 \times (2+1) \\ = \underline{(12)}$$

$$N \rightarrow \frac{N}{\text{spf}[N]} \leq \frac{N}{2}$$

2, 3, 5, 7, 11

$$N \rightarrow \frac{N}{2} \xrightarrow{1/2} \frac{N}{4} \dots \dots \dots 1$$

$$TC \leq \underline{d(\log N)}$$

given spf
 pre-computed

$$\text{spf}[1] \rightarrow \underline{O(N \log(\log N))}$$

H.W

when will calculation spf
 and solving make sense?

